

Layer 5 — Platform orchestration

Product-owner version

Product goal: turn the prepared research engine, search loops, remote execution, and reasoning tools into a managed experiment platform.

Layer 1 prepared runtime.

Layer 2 prepared roles and workspaces.

Layer 3 prepared scientific execution.

Layer 4 prepared reasoning over knowledge and results.

Layer 5 prepares the **orchestration platform**.

Layer 5 answers:

Can I define a GRN/NCA/ART experiment as a structured object?

Can a controller run the right repo command, track status, collect artifacts, handle failures, and resume or compare runs?

Can Paperclip show the human-facing lifecycle of experiments, reports, candidate rankings, failures, and review actions?

Can the full discovery system become repeatable instead of a collection of manual commands?

Layer 5 boundary

Layer 5 should do:

define experiment objects
define controller/runtime contracts
orchestrate repo CLI commands
track status and artifacts

connect Agentfield to the NCA-ART-GRN research engine
connect Paperclip to Agentfield status and actions
prepare future GRN discovery platform workflows

Layer 5 should not do:

not rewrite the research engine
not replace config setup
not replace the Zettelkasten
not become the LaTeX publisher
not hide scientific assumptions
not treat final images as final success
not launch uncontrolled remote jobs
not bypass role/user/environment policy

The core rule is:

config prepares the machine and roles.
nca-art-grn owns scientific execution.
OpenClaw reasons over selected context.
Agentfield orchestrates experiments.
Paperclip exposes the workflow to humans.

Bundles inside Layer 5

Bundle 11 – Agentfield experiment runtime foundation
Bundle 12 – Paperclip-Agentfield adapter
Bundle 13 – Agentic GRN discovery platform

Layer 5 system relationship

config

prepares users, profiles, mounts, Python envs, packages, workspaces, manifests, folders, smoke commands, and safe runtime checks

nca-art-grn

owns DSL, PDE/ODE simulator, NCA, ART2, ARTMAP, search, perturbation tests, reports, and CLI commands

OpenClaw / PKM reasoning

summarizes reports, suggests next experiments, helps with paper/review context

Agentfield

owns experiment objects, lifecycle, controller execution, state, retries, artifact tracking, status, and campaign orchestration

Paperclip-Agentfield adapter

converts Paperclip jobs/actions into Agentfield experiment requests and converts Agentfield status/results back into Paperclip-visible objects

Paperclip

owns the human-facing dashboard, inbox, approvals, review tasks, run overview, governance, and end-to-end visibility

Bundle 11 — Agentfield experiment runtime foundation

Product outcome

A minimal Agentfield runtime that can represent and execute one structured experiment.

This bundle is the first point where Agentfield becomes real.

It should define:

```
GRNExperiment object
experiment spec schema
experiment status schema
controller loop
repo command runner
artifact collector
failure recorder
result summarizer
local/Runpod execution target
```

Product value

Bundle 11 turns manual commands like:

```
python -m nca_art_grn.cli.run --config configs/experiments/smoke_nca_art_pipeline.yaml
```

into structured platform requests like:

```
GRNExperiment:
  method: nca_art_pipeline
  target: local
  config: configs/experiments/smoke_nca_art_pipeline.yaml
  candidate_batch: ...
  expected_artifacts:
    - mechanism_report.md
    - pattern_dynamics.json
    - art2_prototypes.json
    - artmap_transitions.json
```

It does not own the science. It owns the lifecycle around the science.

Concretizations

```
prepare_agentfield_runtime_workspace  
prepare_grn_experiment_schema  
prepare_experiment_status_schema  
prepare_agentfield_controller_skeleton  
prepare_repo_command_runner  
prepare_artifact_collector  
prepare_failure_status_runtime  
prepare_agentfield_local_smoke_config  
check_agentfield_runtime_ready  
run_agentfield_local_smoke
```

What this does for the whole system

Bundle 3/4/5 provide executable research commands.
Bundle 11 wraps those commands in experiment lifecycle state.
Bundle 12 later exposes that lifecycle to Paperclip.
Bundle 13 later uses it for larger autonomous GRN discovery workflows.

Bundle 12 — Paperclip-Agentfield adapter

Product outcome

A bridge between the human-facing Paperclip workflow and Agentfield's experiment runtime.

This bundle answers:

Can Paperclip request an experiment without knowing all repo internals?

Can Paperclip show whether an Agentfield experiment is queued, running, failed, or complete?

Can Paperclip show links to mechanism reports, search reports, prototypes, transition graphs,

figures, candidate DSL files, and failure reasons?

Can a human approve, reject, retry, annotate, or escalate a run?

Product value

This is the visibility and governance layer.

Agentfield can run experiments, but Paperclip makes them human-manageable:

Paperclip job/action

- > adapter
- > Agentfield GRNExperiment
- > nca-art-grn execution
- > Agentfield status/results
- > adapter
- > Paperclip dashboard/review item

Concretizations

```
prepare_paperclip_agentfield_adapter_workspace  
prepare_paperclip_job_schema  
prepare_agentfield_request_mapper  
prepare_agentfield_status_mapper  
prepare_artifact_link_mapper  
prepare_review_action_mapper  
prepare_adapter_smoke_fixtures  
check_paperclip_agentfield_adapter_ready  
run_paperclip_agentfield_adapter_smoke
```

What this does for the whole system

Bundle 11 owns experiment execution state.
Bundle 12 lets Paperclip request and review that state.
Paperclip remains the dashboard.
Agentfield remains the controller.
nca-art-grn remains the scientific engine.

Bundle 13 — Agentic GRN discovery platform

Product outcome

A higher-level GRN discovery workflow that can coordinate candidate generation, mechanism testing, search, remote execution, reasoning, and review.

Bundle 13 is where the system becomes an integrated research platform rather than isolated tools.

It should coordinate:

DSL candidate generation
PDE/ODE simulation
NCA training/evaluation
ART2 prototype discovery
ARTMAP transition learning
prototype-to-DSL mapping
parameter search comparison
robustness sweeps
perturbation design
Runpod execution
mechanism reports
reasoning summaries
Paperclip review tasks

Product value

Bundle 13 turns the earlier bundles into campaigns:

Generate candidate mechanisms.
Evaluate mechanism dynamics.
Train/test NCA local rules.
Discover ART2 prototypes.
Learn ARTMAP transitions.
Map back to DSL.
Run perturbation and robustness tests.
Rank candidates.
Suggest next experiments.
Ask human to approve next campaign.

It should still obey the scientific guardrail:

The goal is not pretty patterns.
The goal is mechanism evidence and experimental design.

Concretizations

prepare_grn_discovery_campaign_schema
prepare_candidate_generation_agent
prepare_mechanism_evaluation_agent
prepare_search_strategy_agent
prepare_perturbation_design_agent
prepare_artifact_review_agent
prepare_next_experiment_agent
prepare_human_review_gate
prepare_campaign_state_store
prepare_grn_discovery_campaign_smoke

check and discover what from needs



What this does for the whole system

Bundle 3 supplies the core mechanism evaluation loop.

Bundle 4 supplies search and comparison.

Bundle 5 supplies remote execution.

Bundle 8 supplies reasoning.

Bundle 11 supplies Agentfield runtime.

Bundle 12 supplies Paperclip visibility.

Bundle 13 composes them into one discovery platform.

Layer 5 role ownership

Layer 5 uses multiple roles, but each role has a clear responsibility.

operator / vmuser

runs config bootstrap/check/smoke steps

manages system-level readiness and safe execution

aiengineer

develops Agentfield, adapter, controller, schemas, and platform logic

researchscientist

owns the scientific configs, experiment meaning, candidate interpretation,
and mechanism validity

publisher

later consumes reviewed reports and summaries for notes and papers

The key split:

AI Engineer builds the orchestration platform.
Research Scientist defines what counts as scientific evidence.
Operator keeps execution safe and repeatable.
Publisher turns reviewed evidence into knowledge and papers.

Layer 5 success condition

After Layer 5, you should be able to say:

I can define a GRN experiment as structured data.
Agentfield can execute it through the nca-art-grn repo.
The run has status, artifacts, failure reasons, and reports.
Paperclip can show the run to a human.
A human can review outputs and approve next actions.
A larger GRN discovery campaign can be composed from smaller tested pieces.

In one line:

Layer 5 turns the prepared research engine into a managed, reviewable, agent-orchestrated discovery platform.

Layer 5 acceptance themes

Layer 5 is acceptable only if:

Agentfield calls existing repo commands instead of duplicating science code.
Paperclip displays experiment lifecycle instead of becoming the controller.
Config remains responsible for setup, roles, envs, and safe bootstrap.
Research artifacts remain under /workspace/runs and /workspace/artifacts.

Mechanism reports remain central outputs.
Human review gates exist before expensive or publication-relevant actions.
Final pattern similarity is never treated as sufficient evidence.

Layer 5 final product view

Before Layer 5:

You have prepared tools, repos, configs, reports, and reasoning workflows.

After Layer 5:

You have a platform that can coordinate those pieces as experiments and campaigns.

The platform stack becomes:

config

machine + role + environment + workspace readiness

nca-art-grn

scientific mechanism engine

OpenClaw / PKM

reasoning and knowledge layer

Agentfield

experiment orchestration layer

Paperclip adapter

integration layer

Paperclip

human-facing control and review layer

